Measurement of $\nu_\mu + \bar{\nu}_\mu$ Charged-Current Multi-Pion Production on Argon with the MicroBooNE Detector in the NuMI Beam

Technical Note: the Two-Charged-Pion Final State (`CC1mu2piBDT`)

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Abstract

We describe the extension of the MicroBooNE NuMI $\nu_\mu + \bar{\nu}_\mu$ charged-current single-charged-pion analysis to final states with two charged pions and any number of nucleons, CC2$\pi^{\pm}Xp$. The selection follows the strategy of the BNB CC2$\pi^{\pm}Np$ analysis [3]: a four-class particle classifier (muon, pion, proton, other) on every track, the assignment of the tracks to one muon and N pions by the largest summed log-probability, and an event classifier on the assigned probabilities, trained in two folds by event number so that no simulated event is scored by a model that saw it. On the full NuMI exposure (7.766×10^{20} POT in forward and 1.1082×10^{21} POT in reverse horn current) it selects 15% of the signal at 48% purity in the combined configuration, against 18% at 26% for the selection of the first draft of this note. Without a constraint the uncertainty of the one-bin total is dominated by the flux, amplified by the subtraction of a background as large as the signal, and by the detector variations. Four control samples defined from the classifier outputs condition the covariance of the extraction framework, partly through the signal they contain: in the combined configuration the one-bin total carries 16.5% with the constraint (19.0% with the MCS-calibration and soft-pion efficiency terms) and 50% without, with data statistics then the largest term; treatments that reduce the transfer of the signal model from the sidebands give 20 to 24%. Ensembles of pseudo-experiments return the expected truth within 1%, with nominal statistical coverage without the constraint and a conservative statistical term with it, and ensemble means for two reweighted interaction models lie within 0.11σ of their own truth. In the opened control regions of the single-pion analysis, calibrating the MCS momentum of exiting tracks as in the single-pion analysis brings the classifier inputs much closer to the beam-on data; with the systematic covariance of the prediction the class probabilities, the MCS momentum and the event score are statistically compatible with the data, while the LLR score remains incompatible at its proton end, where the long protons sit ($p = 0.016$), and follows the recombination variation of the detector term at 1.4 times its size, which moves no result by more than 0.12 of its uncertainty beyond that term. For differential measurements, the muon momentum and angle, the angles of the two pions, their opening angles and the proton multiplicity pass the 0.50 migration criterion with 2 to 8 bins and 8

to 23% data statistics per bin; the pion momenta, limited by the saturation of the range momentum, give 2 or 3 bins. Extracted through the framework with the sideband constraint, the seven variables are compatible with their regularisation-smeared truths in the nominal pseudo-data realisation, with 17 to 59% uncertainty per bin and 17 to 20% on their integrals in the combined configuration; ensembles of pseudo-experiments show nominal statistical coverage per bin and close within 0.43 of the total uncertainty under two reweighted models. Three or more pions are not measurable at this exposure and serve as a control sample. The signal region is blind, and every result is extracted from pseudo-data.

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1 Introduction

Charged-current (CC) production of multiple charged pions on nuclei sits at the transition between the resonant (RES) and deep-inelastic (DIS) regimes of neutrino–nucleus scattering and is strongly shaped by the intranuclear hadronic cascade. A ν_μ interaction that produces several pions at the primary vertex may have some of them absorbed, charge-exchanged or produced in re-scattering before they leave the nucleus, so the observed charged-pion multiplicity is a direct probe of final-state interactions (FSI). Multi-pion final states are also a background to exclusive single-pion and pion-less measurements, so a measurement of their rate on argon is useful both as a cross-section result and as a constraint on the interaction model used across the MicroBooNE programme.

This note documents the extension of the NuMI $\nu_\mu + \bar{\nu}_\mu$ CC1$\pi^\pm$ analysis [4] to two charged pions. The first draft of this note (2026-08-13) defined the selections `CC1mu2pi` and `CC1mu3pi` as subselections of the single-pion selection `CC1mu1piXp` with a looser pion identification [1, 2], on the Run-1 forward-horn overlay only. Since then the analysis has been rebuilt on the strategy of the BNB CC2$\pi^\pm Np$ analysis [3] and on the current single-pion release: a baseline on the full exposure (Section 4), a particle classifier (Section 5), the track assignment and an event classifier (Section 6), a study of the detector term (Section 7), control samples and a background constraint (Section 8), the extraction of the one-bin total through the framework (Section 9), and the comparison of its inputs with beam-on data in the opened control regions, with the calibration of the MCS momentum and the selection variants (Section 10), the two-fold event classifier that removes an overtraining of the first one (Section 11), and the variables of the differential measurements (Section 12). The samples, normalisation, systematic universes and Wiener-SVD unfolding [10] are those of the single-pion analysis.

The decisions that fix the scope are:

- D1: the signal is inclusive in nucleons, CC2$\pi^\pm Xp$, with a subsample with at least one proton for comparison with the BNB analysis.
- D2: a per-pion momentum threshold of 0.10 GeV/c for the signal, totals and angles; pion-momentum bins only above 0.175 GeV/c, where the range estimator is unbiased.
- D3: the two-pion signal region is blind. Part of it lies in the multi-π control region of the single-pion analysis, which has been compared with beam-on data.
- D4: NuGraph inputs are not used; NuMI samples processed with NuGraph are not available.
- D5: an independent-generator closure with a NuMI NuWro overlay, conditional on the validation of that sample.
- D6: the truth requirement $\theta(\mu, \text{leading } \pi) < 2.6$ rad has a reconstructed counterpart on the longer assigned pion, the track length standing in for the momentum.
- D7: no exclusivity cut against the blind single-pion signal region; the overlapping events are opened only after the single-pion signal region.
- D8: the measurements stop at two pions. Events with three or more pion candidates are a control sample of the two-pion background.

2 Signal definition

The signal is a ν_μ or $\bar{\nu}_\mu$ charged-current interaction with its vertex in the fiducial volume of the single-pion analysis, one muon with $p_\mu > 0.15$ GeV/c, exactly two charged pions leaving the nucleus, no neutral pion, kaon or heavier meson, and any number of nucleons. The pion charge is not measured. Both pions must have $p_\pi > 0.10$ GeV/c; the multiplicity is counted at any momentum, so an event with a third charged pion below 0.10 GeV/c is not signal (such events amount to 6.2% of the signal on the Run-4 forward-horn overlay). The opening angle between the muon and the leading pion, the one of highest true momentum, is below 2.6 rad, as in the single-pion definition. The subsample with at least one proton above 0.3 GeV/c is 81% of the

signal.

Pion-momentum threshold. In a two- or three-pion event the sub-leading pions are softer than the leading one, and the softest pion sets how often the full topology can be reconstructed. On the Run-1 forward-horn overlay the mean leading (softest) true pion momentum is 0.52 (0.27) GeV/c for two pions and 0.57 (0.21) GeV/c for three. The efficiency of the first-draft selection, evaluated on the signal whose softest pion lies above a threshold $p_\pi^{\min}$, is given in Table 1.

Table 1: Signal efficiency of the first-draft selections as a function of a per-pion momentum threshold $p_\pi^{\min}$ applied to the softest pion, Run-1 forward-horn overlay. "Kept" is the fraction of the $p_\pi > 0$ signal retained; Eff. is the efficiency within the restricted phase space.

	Two pions		Three pions	
$p_\pi^{\min}$ [GeV/c]	Kept	Eff. [%]	Kept	Eff. [%]
0	1.00	16.3	1.00	6.4
0.10	0.90	17.1	0.82	6.2
0.175	0.70	18.9	0.49	8.1
0.25	0.43	20.2	0.26	5.6

Raising the threshold changes the efficiency little: with every pion above 0.175 GeV/c about four in five events are still lost, so the efficiency is limited by the reconstruction of several tracks from one vertex, not by the soft pions. The threshold of 0.10 GeV/c, the pion tracking turn-on, removes the unreconstructable tail (10% of the two-pion signal) and keeps 90% of it; 0.175 GeV/c would keep 65% of the two-pion signal in the combined configuration. Pion-momentum bins start at 0.175 GeV/c (D2).

3 Samples, exposure and normalisation

The samples and normalisation are those of the single-pion release: fourteen ν_μ overlay files (forward horn current Runs 1, 2, 4, 5; reverse horn current Runs 1, 2, 3, 4a, 4b, 4c), seven beam-off and seven dirt samples, each run normalised to its recorded exposure, 7.766×10^{20} POT in forward horn current (FHC), 1.1082×10^{21} POT in reverse horn current (RHC) and 1.8848×10^{21} POT combined. The detector variations are the Run-4 FHC and RHC samples, a central-value sample and eight variations each (light yield down and Rayleigh scattering, recombination, space charge and four wire-response variations). The beam-on data are not used: the pseudo-data are Poisson throws, run by run, of the overlays and the dirt at the data exposure, to which the framework adds the beam-off prediction. Generator predictions with the two-pion signal definition come from the standalone GENIE, GiBUU, NuWro and NEUT samples of the single-pion analysis (Section 12).

4 Baseline on the full exposure

`CC1mu2pi` and `CC1mu3pi` inherit the `CC1mu1piXp` selection (slice, fiducial volume, topology and muon candidate) with a looser pion identification: vertex distance up to 9.5 cm, the pion BDT of the first draft in place of the single-pion identification, up to two uncontained pion candidates, no shower veto and no single-pion opening-angle cut. Following D6 they now require $\theta(\mu, \text{longest pion candidate}) < 2.6$ rad. The selections store every counted candidate (containment, length, range and MCS momenta, direction, identification scores and the backtracked truth), the true pions, the proton and kaon counts and a truth topology class. On a 30 000-event

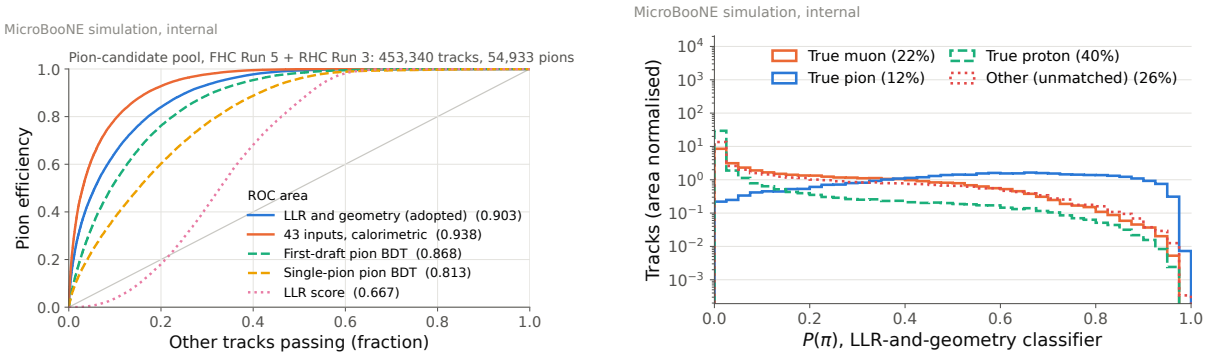


Figure 1: Left: pion against everything else on the pion-candidate pool of the held-out runs, unmatched tracks as "other" (the areas of Table 3 are for tracks with a truth match). Right: $P(\pi)$ of the LLR-and-geometry classifier by true class, same tracks, area-normalised.

slice every branch of the single-pion selection is identical before and after these changes. Table 2 gives the baseline at the full exposure, beam-off and dirt included.

Table 2: Baseline of the first-draft selections at the full exposure, with the reconstructed opening-angle requirement. Purities include beam-off and dirt.

	Generated signal	Selected signal	Selected total	Efficiency	Purity
Two pions, FHC	1 210	221	836	18.2%	26.4%
Two pions, RHC	1 672	303	1 168	18.1%	26.0%
Two pions, combined	2 882	524	2 004	18.2%	26.1%
Three pions, FHC	204	21.7	203	10.6%	10.7%
Three pions, RHC	347	32.8	314	9.5%	10.4%
Three pions, combined	551	54.5	517	9.9%	10.5%

These numbers replace the raw Run-1 counts of the first draft, whose "670 (66) events at full exposure" were overlay counts at the simulated exposure. The largest two-pion backgrounds are single-pion events with a second counted candidate (22% of the sample), events with a neutral pion (21%), three or more charged pions (7%), neutral-current events (6%) and beam-off (2.5%). In selected signal events only 60% have both counted candidates backtracked to true pions, and the longest candidate is the leading pion in 52%, so observables ordered by pion cannot be used with this candidate choice. The three-pion sample is 10.5% pure and its largest background, 30%, is a two-pion event with a third counted candidate.

5 Particle classifier

A four-class classifier (muon, pion, proton, other) is trained with XGBoost [5] on the tracks of the pion-candidate pool of the selection (generation 2, track score ≥ 0.3), dumped from all fourteen overlays (3.1 million tracks). Tracks are labelled by their backtracked particle; tracks without a truth match, mostly cosmic rays of the overlay, are the class "other". Training uses run periods 1, 2 and 4 of both horn modes; the held-out periods are RHC Run 3 and FHC Run 5. The beam-on, beam-off and dirt ntuples of the older production (Runs 1–3) lack twenty of the track branches every overlay carries (the Bragg-pion likelihoods, truncated dE/dx, deflection, end spacepoints, hit and descendant counts), so the deployed models use the 43 inputs every sample carries. The model is evaluated in the selection through ROOT's `RBDT` [6], which reproduces XGBoost exactly. Figure 1 shows the pion ROC curves on the whole pool of the held-out runs and the pion probability of the adopted classifier by true class.

Table 3: ROC area of pion identifications on the two-pion candidate pool, held-out runs, tracks with a truth match (backtracked purity above 0.5).

Pion identification	π vs rest	π vs p	π vs μ
Particle classifier (63 inputs)	0.954	0.985	0.895
Pion BDT of the first draft, as applied	0.888	0.945	0.804
Pion BDT of the single-pion analysis	0.818	0.961	0.558
MIP BDT	0.718	0.957	0.271
LLR score	0.670	0.893	0.251

The gain is the same in FHC and RHC and for π^+ and π^-. With the unmatched tracks as the fourth class, at the pion efficiency of the first-draft BDT (91.8%) the purity of the counted pion candidates rises from 27.6% to 43.3% and the fraction of protons passing falls from 13.4% to 4.1% (63 inputs). With the 43 inputs every sample carries, the pion ROC area on the full pool, unmatched tracks included, is 0.938 (0.946 with all inputs, 0.868 for the first-draft BDT). The pion efficiency does not depend on the true pion momentum more than that of the first-draft BDT and is higher at threshold. Muon against pion is the hardest separation (0.895; about a third of the true muons in the pool still pass).

Used as the pion identification of the single-pion selection at the same pion efficiency, the 43-input classifier raises the purity of the counted tracks from 52.6% to 64.8% and, at event level on the held-out runs, the selection purity from 58.0% to 64.7% for 5% of the signal. This is a study for a later revision of the single-pion analysis.

The LLR-and-geometry classifier. Section 7 shows that the detector variations act on the calorimetric inputs. A second model drops the Bragg likelihoods, χ^2 scores, PIDA and calorimetric energies, and keeps the LLR score (summed and per plane), length, track score, vertex distance, start and end containment, range and MCS momenta, the proton-hypothesis energy, the daughter counts and the per-plane hit flags. Its pion ROC area is 0.892 against 0.926 for the deployed model on the same tracks, and at 80% pion efficiency it passes 9.0% of the protons against 3.9%. It is the particle classifier of `CC1mu2piBDT`.

6 Track assignment and event classifier

Sections 6 to 10 develop the selection with a single event classifier, which Section 11 replaces by the two-fold one of the final selection.

Assignment. In events with one neutrino slice and the vertex in the fiducial volume, the tracks of the pion pool with track score ≥ 0.5, the muon candidate included, carry the four class probabilities. For the hypothesis "one muon, N pions, the other tracks protons or other" the log-likelihood is

$$L_N = \log P_\mu(\mu) + \sum_{\text{pions}} \log P_\pi + \sum_{\text{rest}} \log\left(1 - P_\mu - P_\pi\right), \tag{1}$$

where $1 - P_\mu - P_\pi = P_p + P_{\text{other}}$ and every probability is floored at 10^{-6} before the logarithm; every choice of the muon and the N pions among at most eight tracks is enumerated, and the best and the runner-up are kept for $N = 1, 2, 3$. In the selected signal of the held-out periods the assigned muon is a true muon in 89% of events and each assigned pion a true charged pion in 88%; all three are in 76%. In the preselected signal, however, only 30% of the events have the muon and both pions as tracks, so in most signal events no assignment can be fully right.

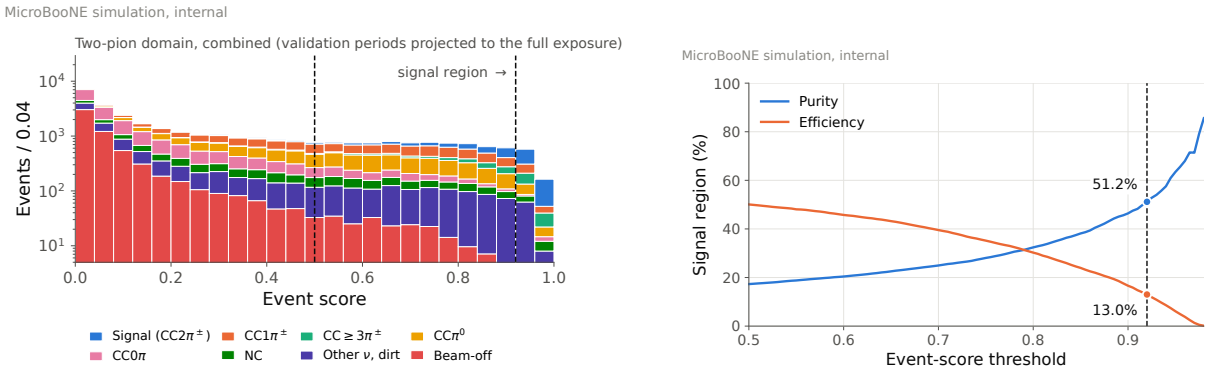


Figure 2: Event score of the single event classifier (LLR-and-geometry particle classifier, MCS-calibrated inputs) in the two-pion domain, validation periods projected to the full exposure, combined configuration (left), and the purity and efficiency of the signal region against the threshold (right).

Event classifier. An XGBoost classifier separates the two-pion signal from everything else on 33 features: the probabilities of the assigned tracks, $L_2 - L_1$, $L_2 - L_3$, the margin over the runner-up, the class sums and the counts above 0.5 over all tracks, the track and shower counts, the topological score and CosmicIP. Its domain is one slice, the vertex in the fiducial volume, the software trigger, at least three tracks and $\theta(\mu, \text{longest assigned pion}) < 2.6$ rad (D6). It is trained on run periods 1, 2 and 4 of both horn modes (206 152 events, 13 558 signal) with the nominal mixture of overlay, dirt and beam-off and the central-value weights, and tested on FHC Run 5 and RHC Run 3 (104 814 events, 7 253 signal), which the particle classifier did not see either. These periods also choose the working point below, so they are validation periods rather than an untouched test sample. The ROC area on them is 0.887; adding the lengths, vertex distances, containment and opening angles of the assigned tracks raises it to 0.898, a gain that disappears in the total uncertainty, so these kinematic features are not used. The largest gains come from the summed pion probability, L_2, the number of tracks with $P_\pi > 0.5$, the topological score and the number with $P_\mu > 0.5$. Figure 2 shows the score of the LLR-and-geometry selection by true class.

Working point. The threshold is chosen on the expected total uncertainty of the one-bin total, with the validation-period yields projected to the full exposure: data statistics, flux ($17\% \times (1 + B/S)$), POT and targets, cross section, re-interaction, detector (the eight Run-4 variations, FHC and RHC correlated) and simulation statistics (Table 4). With statistics and flux alone the minimum lies at 3.9% efficiency; the detector term rises steeply above a score of 0.95, and with all terms the minimum is at 0.92 (with the two-fold classifier of Section 11, 0.90), flat between 0.90 and 0.94 within the statistics of the detector-variation samples (Figure 3).

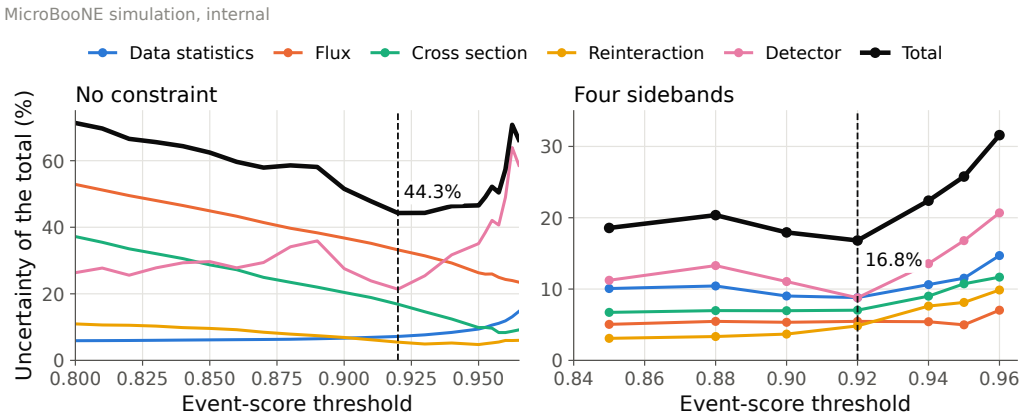


Figure 3: Expected uncertainty of the combined one-bin total against the score threshold: without a constraint (flux 17% $\times$ $(1 + B/S)$, left) and with the four sidebands (Section 8, right).

Table 4: Expected uncertainty of the one-bin total (combined configuration) for the first-draft selection, the same with the particle classifier as pion identification (`CC1mu2piNewPID`), and the event classifier with the deployed and with the LLR-and-geometry particle classifier at score > 0.92. The last row is the evaluation of Section 8, with the flux from the PPFX universes [9] instead of 17% $\times$ $(1 + B/S)$. The totals also contain the POT, target and simulation-statistics terms, which are not listed.

Selection	Eff.	Purity	Stat.	Flux	Cross sect.	Reint.	Detector	Total
`CC1mu2pi`	18.2%	26%	8%	65%	47%	17%	34%	89%
`CC1mu2piNewPID`	15.8%	35%	8%	48%	27%	13%	21%	61%
Event classifier, deployed	17.1%	56%	6%	31%	12%	5%	33%	47%
Event classifier, LLR and geometry	13.0%	51%	7%	35%	17%	6%	21%	46%

The cross-section term is the model dependence of the background and falls with the purity; the model dependence of the efficiency within the GENIE universes stays below 6%. In the combined configuration the signal region of the LLR-and-geometry selection holds 738 predicted events at full exposure, 378 of them signal; the largest backgrounds are single-pion events (15% of the sample) and three or more charged pions (13%). Of the selected events 14% lie in the blind single-pion signal region (D7) and 12% in the opened multi-π control region of the single-pion analysis (D3).

Selection requirements. Table 5 lists the requirements of `CC1mu2piBDT`. The cuts of the single-pion selection after its preselection (topological score above 0.67 and the muon-candidate, pion-containment, wire-gap, shower and track-multiplicity requirements) are not applied; the topological score, CosmicIP and the track and shower counts enter only as features of the event classifier.

Table 5: Final requirements of `CC1mu2piBDT`, with the two-fold event classifier and the signal-region threshold 0.90 (Section 11); Sections 6 to 10 document the development with the superseded single classifier and threshold 0.92. Cuts 1–9 define the domain of the event classifier; the signal region adds cut 10. The sidebands (Section 8) are the events that pass cuts 1–9 and fail cut 10, assigned in the order listed; in the extraction they exclude the events selected by the single-pion analysis. The proton, muon and MIP Bragg likelihoods of cut 5 exist in every production, unlike the pion one, and serve only as a validity requirement of the pool; they are not classifier inputs.

	Cut	Requirement
Event		
1	Neutrino slices	exactly one
2	Reconstructed vertex	inside the fiducial volume of the single-pion analysis, outside the dead-wire region in z
3	Software trigger	passed
Track pool: direct neutrino daughters		
4	Track score	≥ 0.5
5	PID inputs	LLR score in $(-1, 2)$; proton, muon and MIP Bragg likelihoods in $(0, 500)$; length > 0
6	Ranking	by $\max(P_\mu, P_\pi)$ of the LLR-and-geometry particle classifier; at most eight tracks kept
7	Pool size	at least three tracks
Assignment and event classifier		
8	Assignment	the muon and two pions that maximise L_2
9	Opening angle	$\theta(\mu, \text{longer assigned pion}) < 2.6$ rad (D6)
10	Event score	> 0.90 (model of the other event-number parity): signal region
Sidebands: cuts 1–9, score ≤ 0.90		
	Three-pion-like	at least four tracks and $L_3 > L_2$
	Proton-like	an assigned pion with $P_p > 0.5$
	Score band	$0.5 <$ score ≤ 0.90
	Low score	score ≤ 0.5

7 The detector term

At the working point of the deployed classifier the detector term, 33%, is the largest. A paired bootstrap over the physical events, with one Poisson weight per event shared by the central-value sample and the eight variations, attributes 7.7% of it to the statistics of the variation samples. It comes from the recombination (+22%) and wire-response variations (angle yz +18%, x +12%, yz +8%); the light-yield, space-charge and angle-xz variations are consistent with zero. At track level these variations move the χ^2 scores, PIDA, the Bragg likelihoods and the calorimetric energies by up to half of their spread for protons, the LLR score by at most 0.04 and the geometric inputs not at all. At event level they lower the selected signal by about 8% and the background by 10–18%, and the detector term splits into an efficiency part (14.3%), which no background constraint can remove, and a background part (17.8%).

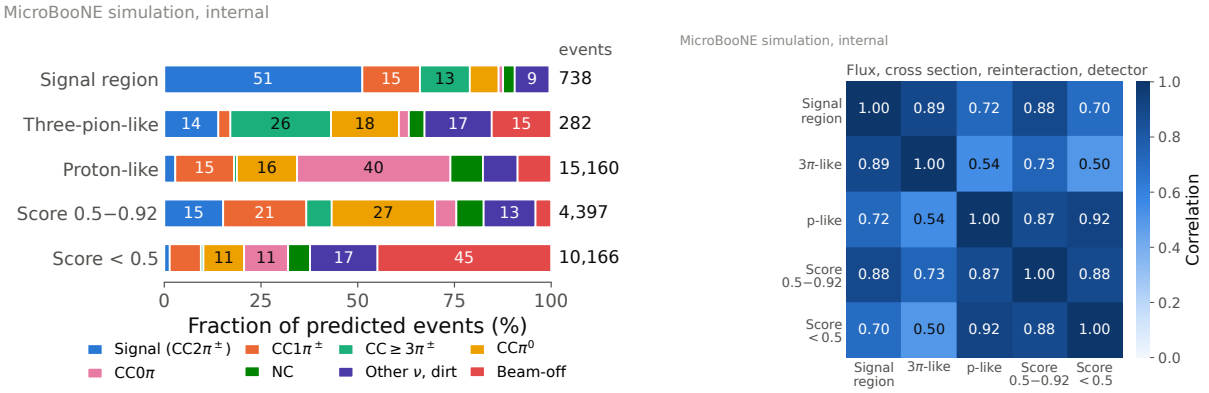


Figure 4: Predicted composition of the signal region and the sidebands (left) and the correlation of their predicted counts under the flux, cross-section, reinteraction and detector covariance (right), combined configuration, full exposure.

Table 6: Variants of the particle classifier and of the event features (combined configuration), at the best threshold of each. The detector term is split into its efficiency and background parts. The flux term is 17% × $(1 + B/S)$, as in the scans of Table 4; with the PPFX universes the LLR-and-geometry selection has 45.5% (Table 8), the 46% of Table 4.

Particle classifier	Event features	Score >	Eff.	Purity	Detector	(eff., bkg.)	Total
Deployed	PID set	0.92	17.1%	56%	32.7%	(14.3, 17.8)	47%
Deployed	full set	0.94	14.4%	62%	33.0%	(18.2, 14.8)	45%
Without χ^2, PIDA, energy	PID set	0.93	14.4%	57%	35.1%	(14.9, 19.0)	48%
LLR and geometry	PID set	0.92	13.0%	51%	21.4%	(7.8, 15.2)	44%
LLR and geometry	full set	0.93	12.0%	55%	20.3%	(8.4, 12.9)	42%

No variant removes the detector term, and the totals lie within the 3% scatter of the scans. The LLR-and-geometry classifier halves the efficiency part (14.3% to 7.8%), the part a constraint cannot reach, and is adopted. Training the event classifier or the particle classifier with the variation samples did not help.

For comparison, the BNB analysis quotes a detector term of 11.0% for the predicted selected event count, from seven Run-4d variations without the wire-angle ones [3]. In the same convention the LLR-and-geometry selection has 10.6% with all eight variations and 8.4% with those seven; the terms of Table 4 are those of the extracted total, in which the background subtraction and the efficiency correction amplify the change of the predicted events by a factor of two to four. The recombination variation leads in both analyses.

8 Control samples and the background constraint

The flux term scales the signal and the background together, and the background is as large as the signal. Control samples that share the background composition constrain it through the conditional covariance of the extraction framework (`ConstrainedCalculator`): the covariance of the predicted counts in the signal region and the sidebands, built from every source, is conditioned on the sidebands, and the background estimate moves with the sideband data. Four sidebands are defined from the classifier outputs, exclusive and in this order inside the two-pion domain (Table 5): three-pion-like (at least four tracks and three pions more likely than two; the control sample of D8), proton-like (an assigned pion with $P_p > 0.5$), event score between 0.5 and the working point, and event score below 0.5. The sidebands leave out the events of the blind single-pion signal region, so they can be opened without it (Table 7, Figure 4).

Table 7: Signal region and sidebands of the LLR-and-geometry selection with the single event classifier (score > 0.92; Table 12 for the two-fold one), combined configuration, predicted events at full exposure.

Region	Events	Signal	Purity	Largest classes
Signal region (score > 0.92)	738	378	51%	CC1π 15%, CC$\geq 3\pi$ 13%
Three-pion-like	282	39	14%	CC$\geq 3\pi$ 26%, CCπ^0 18%, beam-off 15%
Proton-like	15 160	431	3%	CC0π 40%, CCπ^0 16%, CC1π 15%
Score 0.5–0.92	4 397	669	15%	CCπ^0 27%, CC1π 21%
Score < 0.5	10 166	143	1%	beam-off 45%, CC0π 11%, CCπ^0 11%

A first evaluation outside the framework, with the formalism of `ConstrainedCalculator`, gives the uncertainties of Table 8. The flux term falls from 35% to 5.5%, since the flux scales the background and the sidebands alike, and the detector term from 21% to 9%, close to its efficiency part. The three-pion-like sample alone, 282 events, halves the total. The data-statistics term grows with a constraint, because the counts of the sidebands enter the background estimate. A single sideband whose composition differs from the background of the signal region can raise the detector term: the sideband below a score of 0.5 is 45% beam-off, which the detector variations do not move, and the proton-like one 40% CC0π, so the conditioning transfers their detector shifts to the signal region with the wrong weight.

Table 8: Uncertainty of the one-bin total (combined configuration, single event classifier) with each sideband constraint, evaluated outside the framework (flux from the PPFX universes, detector taken fractionally).

Constraint	Total	Flux	Cross sect.	Reint.	Detector	Data stat.	MC stat.
None	45.5%	35.4%	16.7%	5.5%	20.6%	7.2%	3.9%
Three-pion-like	24.2%	13.6%	8.3%	4.2%	11.7%	11.4%	6.8%
Proton-like	31.9%	13.0%	15.2%	5.3%	22.9%	7.3%	4.0%
Score 0.5–0.92	22.5%	6.2%	8.4%	5.2%	17.2%	7.7%	4.2%
Score < 0.5	33.0%	13.6%	10.6%	4.9%	26.2%	7.7%	4.4%
All sidebands	16.8%	5.5%	7.0%	4.8%	8.8%	8.8%	5.0%

Closure under alternative models. Pseudo-data from a reweighted interaction model in every region, extracted with the nominal model, return the alternative truth: with all sidebands 0.952 (-0.29σ) under GENIE [7, 8] multisim universe 545 and 0.994 (-0.03σ) under the $\Delta \to N\pi$ angular variation, against 0.979 and 1.012 without the constraint; over the GENIE multisim universes the bias has an rms of 4.5% with the constraint, within 1σ in every universe, and 17.4% without (98% of the universes within 1σ). These models lie inside the variations that build the covariance, so the test checks the procedure, not the model space.

Signal in the sidebands. The score band holds more predicted signal than the signal region (669 against 378 events), and the variations that move the signal move it in every region together, so the conditioning also carries the nominal signal model from the sidebands into the signal region. Treated as an independent source, uncorrelated with the signal region and the backgrounds, so that the sidebands constrain the background only, the signal variations of the sidebands raise the constrained total from 16.8% to 19.3% (flux 5.5% to 9.6%, cross section 7.0% to 8.6%); leaving the score band out of the constraint gives 19.0%, and both together 21.4%. The score band moves the extraction under universe 545 from 1.012 to 0.952; every variant closes within 0.3σ for both models. The nominal treatment keeps the signal covariance between the regions because the same flux and interaction parameters act coherently on one CC2π process in all of them;

the decorrelated variants are conservative stress tests, not equally plausible covariance models. The two alternative models hardly change the ratio of score-band to signal-region signal (1.684 nominal, 1.660 under universe 545, 1.681 under the $\Delta \to N\pi$ variation), so they do not test this transfer; a stress model outside the covariance does (Section 10).

9 Extraction through the framework

`CC1mu2piBDT` implements the selection in the framework: the track pool, the class probabilities of the LLR-and-geometry classifier, the assignment, the 33 features, the event score (`RBDT`) and the region; the selected events are those of the signal region. On a 30 000-event slice the features equal those of the Python study exactly, the scores to 6×10^{-8} and the regions in every event, and `CC1mu1piXp` is bitwise unchanged. The 28 release samples and the 18 detector-variation samples were processed with it. The bin configuration has the signal region as an ordinary bin and the four sidebands as sideband bins in the signal block, so that their signal is predicted. The keyword `Calculator Constrained` in the extraction configuration selects `ConstrainedCalculator`; the extractor propagates the ordinary block of each covariance and, with the constraint, the part that remains after the conditioning, so that the terms by source add up to the constrained total.

Table 9: One-bin two-pion total through the framework with the single event classifier, on pseudo-data, without and with the sideband constraint; detector variations taken fractionally and weighted to the current flux (Section 9.1). The last column is the extracted value over the truth of the pseudo-data. The total also contains the POT and target and the simulation-statistics terms, which are not listed.

	Total	Detector	Flux	Cross sect.	Reint.	Data stat.	Unfolded/truth
FHC, no constraint	54.0%	34.2%	37.0%	14.7%	5.2%	10.5%	1.056
FHC, constrained	22.5%	11.4%	6.1%	8.7%	5.9%	14.0%	1.020
RHC, no constraint	41.1%	18.1%	31.5%	15.5%	5.4%	8.7%	1.035
RHC, constrained	16.3%	7.3%	4.9%	5.9%	4.1%	11.1%	1.020
Combined, no constraint	46.0%	25.9%	33.7%	15.0%	5.2%	6.7%	1.044
Combined, constrained	15.7%	8.2%	4.8%	6.6%	5.0%	8.8%	1.021

The constrained combined total, 15.7%, agrees with the evaluation outside the framework (16.8%). With the constraint the data statistics become the largest term, and in the three configurations the pseudo-data realisation returns 1.020–1.021 times its truth.

9.1 Detector-variation samples

The first framework extraction gave a larger detector term than the study outside it, most in RHC (28% against 17%). Per POT, the Run-4 detector-variation central-value samples predict 1.27 (FHC) and 1.51 (RHC) times the event rate of the nominal Run-4 overlays in the fiducial volume, the ratio rising from 1.1 below 0.5 GeV to 1.8–3 above 5 GeV. The samples were simulated with the earlier NuMI flux, the nominal overlays with the current one (Geant v4.10.4): weighted event by event with the ratio of the two fluxes in true energy and angle to the beamline, they predict 0.94 times the nominal rate, flat in energy; the origin of this flat offset is not identified, and the fractional treatment below cancels it. The framework built the detector covariance from the absolute difference between each variation and the variation central value, and so carried that ratio. The covariance type `DVfrac` applies the fractional change of each variation relative to the variation central value to the nominal prediction; it is used here and, since version 1.8, in the single-pion release.

Because the earlier flux overweights high energies, where the two-pion signal sits, the variation samples are in addition weighted to the current flux: the selection framework applies an optional

per-event weight branch, carried only by reweighted copies of the 18 variation samples. On the fractional detector change of the selected two-pion count the weight raises the FHC term from 17.6% to 20.6% and leaves RHC unchanged (9.3% to 9.1%); for the single-pion selection it changes the term by at most 0.3% absolute. Table 10 shows the three treatments.

Table 10: Detector term and total of the one-bin two-pion total for the three treatments of the detector variations.

	Absolute		Fractional		Fractional, flux-weighted	
	Detector	Total	Detector	Total	Detector	Total
FHC, no constraint	40.8%	58.4%	29.8%	51.3%	34.2%	54.0%
FHC, constrained	15.4%	24.9%	11.9%	22.4%	11.4%	22.5%
RHC, no constraint	28.4%	46.6%	17.9%	41.0%	18.1%	41.1%
RHC, constrained	11.1%	18.5%	7.6%	16.3%	7.3%	16.3%
Combined, no constraint	33.4%	50.7%	23.0%	44.5%	25.9%	46.0%
Combined, constrained	11.9%	17.9%	8.5%	15.7%	8.2%	15.7%

The sideband constraint removes the difference between the fractional treatments: the constrained totals are the same with and without the flux weight. The ratio of a variation to the variation CV, which the fractional treatment transfers to the nominal prediction, changes by at most 0.027 (median 0.003) with the flux weight, for signal and background in the signal region and the score band; above 1 GeV it follows the integral within the statistics of the variation samples (5–6%), the largest trend being the recombination variation of the FHC background (0.76 at 1–2 GeV, 1.01 above 5 GeV).

10 Beam-on data in the control regions

The inputs and the output of the two-pion selection were compared with beam-on data in the four control regions of the single-pion analysis, which are open. The skim keeps only events in those regions and outside the signal regions of both selections, so the two-pion signal region is absent from data and prediction alike. The prediction is the sum of the overlays, beam-off data and dirt; the shape tests remove the normalisation and are made per run period and summed. In the two-pion domain the data exceed the prediction by 10–24% (three-pion-like 1.18, proton-like 1.10, score 0.5–0.92 1.20, score below 0.5 1.24), as outside it (1.12); the largest excess is in the π^0 control region.

MCS momentum. The MCS momentum of the pool tracks is the most mis-modelled input: data over prediction rises to 1.14 at 0.7–0.9 GeV/c and falls to 0.39 at 1.8–1.9 GeV/c (shape $\chi^2 = 2234/19$). This is the effect found for exiting muons in the single-pion analysis [4], an angular noise of reconstructed tracks on the scale of the MCS segments that the simulation lacks, calibrated there from version 1.9 by a quantile map of the MCS momentum in cells of the drift-direction component and the length of the track. The same map is applied here to the classifier inputs of every simulated track whose end leaves the containment volume (`CC1MU1PIXP_MP_MCS_MAP`; without it every output is unchanged, bit for bit). With it the shape χ^2 of the MCS momentum falls to 71.5/19 (data over prediction 0.87–1.04, Figure 5), that of $P(\mu)$ from 1100 to 293, of $P(\pi)$ from 256 to 56 (19 degrees of freedom) and of the summed $P(\pi)$ of the event from 92 to 52 (15), in every period.

LLR of long protons. Among pool tracks longer than 40 cm, those with an LLR score below -0.9 (an LLR sum below -630) are 11 times more frequent in the data than predicted (592 against 52); between -0.9 and -0.8 the data hold 1.33 times the prediction and between -0.8

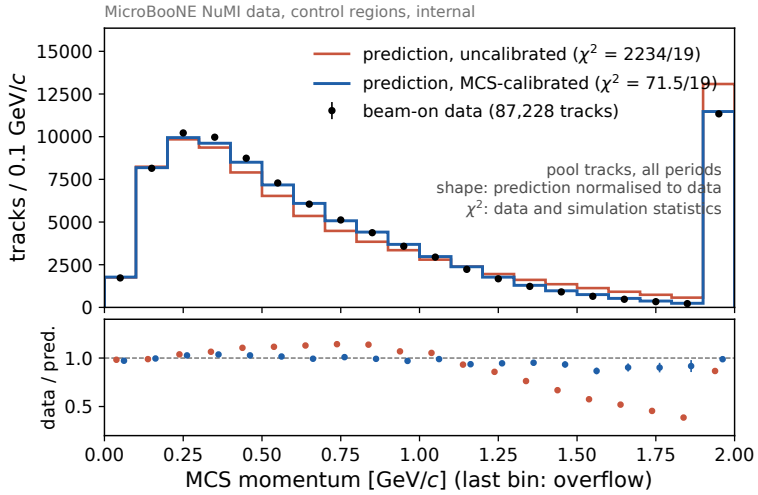


Figure 5: MCS momentum of the pool tracks in the opened control regions, all periods: beam-on data against the prediction (overlays, beam-off, dirt) before and after the MCS calibration, normalised to the data.

and −0.5 0.84. They are long contained protons: both ends contained (97%), starting at the vertex, 40 to 100 cm long, true protons in 72% of the predicted ones, in every period and both productions and on all three planes. With the same number of hits they are more proton-like per hit in the data: for contained tracks longer than 40 cm with an LLR score below −0.5 the median LLR sum is −436 to −486 in the Run-4 data and −339 in the simulation. Of the detector variations the recombination variation reproduces it (median −445; share beyond −0.9 of 0.15, against 0.16–0.23 in the data and 0.015 in the simulation), and the wire-response variations move it the other way. The effect is therefore inside the detector term, of which the recombination variation is the largest part (Section 7, Figure 6). The particle classifier calls these tracks protons in either case: contained tracks longer than 40 cm with an LLR score below −0.5 have $P_p > 0.5$ in 86 to 92% of cases in every LLR band, in data and simulation alike within 1 to 2%.

Event score. With the calibrated simulation the shape χ^2 of the event score is 140/22 (160/22 before). The bin below the cut holds 0.83 of the predicted share, and the recombination variation gives 0.86 in both modes. The lowest bin (0.78) holds control-region CC0π events (97.5%); against the normalisation of that region alone it is at 0.85, the rest coming from the larger excess of the π^0 region in the common normalisation, and the recombination variation gives 0.93–0.94.

With the systematic covariance. The shape tests above carry the data and simulation statistics only. With the covariance of the prediction (flux, cross-section and reinteraction universes, the eight detector variations taken fractionally per horn mode with the statistics of the variation samples inside the detector term, simulation statistics), built from the ingredients of the extraction, the prediction is scaled to the data total, which removes one degree of freedom: empty bins are removed, the last remaining bin is dropped and χ^2 is solved by least squares over the others. The event score gives $\chi^2 = 6.5/11$, the summed $P(\pi)$ 2.0/7, the track $P(\mu)$ and $P(\pi)$ 4.2/9 each and the MCS momentum 6.2/8, against 98.5, 23.6, 48.1, 40.4 and 65.2 with statistics alone in the same, coarser bins (Figure 7). The LLR score of the pool tracks remains incompatible with the data, 20.8/9 ($p = 0.014$), at its proton end, where the long protons sit; the detector term alone brings it from 398 to 36. All tested observables are thus compatible with the prediction within the systematic covariance except the LLR score.

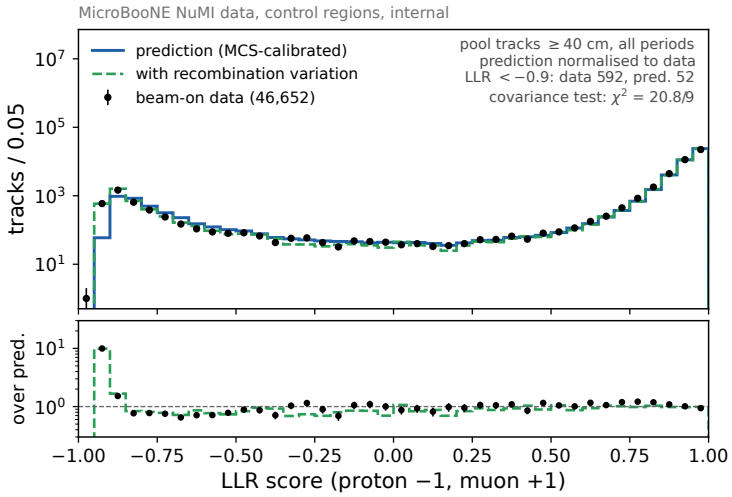


Figure 6: LLR score of pool tracks longer than 40 cm in the opened control regions: beam-on data, the calibrated prediction and the prediction with the overlay changed by the recombination variation, normalised to the data.

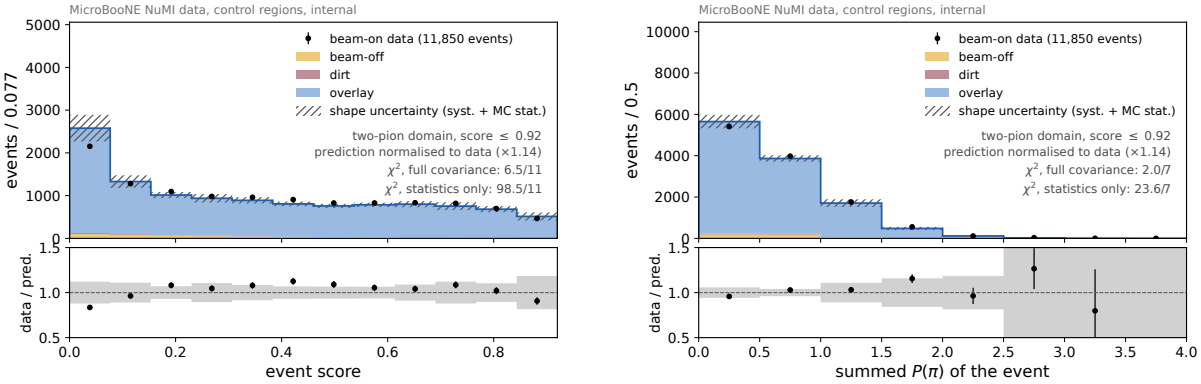


Figure 7: Event score (left) and summed $P(\pi)$ (right) of the two-pion domain in the opened control regions: beam-on data against the calibrated prediction normalised to the data, with the shape uncertainty of the systematic covariance.

Extraction with the calibrated simulation. The overlays, the dirt and the detector-variation samples were processed again with the calibrated classifier inputs, and the pseudo-data thrown with the same seeds, so that they hold the same events. Table 11 gives the totals. The calibration changes the expected content of the signal region by less than 1% (signal −0.9% in FHC and −0.5% in RHC, all events −0.7% and 0.0%) and the constrained combined total from 15.7% to 15.2%. In the combined configuration this measures the flux-averaged CC2$\pi^{\pm}$ cross section to 15% from about 380 selected signal events, the precision of a count of 380 ± 57, of which the data statistics contribute 8.7%. The classification moves about 5% of the signal-region events in or out, and their pseudo-data counts are Poisson draws made without regard to it; this moves the closure to 1.05–1.09, by +0.9 (FHC) and +2.3 (RHC) times the Poisson spread of the migrating events. The calibration uncertainty is evaluated as in the single-pion analysis: pseudo-data with the classifier inputs calibrated by each of the seven alternative maps, extracted against the response calibrated with the nominal map, in four fully correlated terms: 2.2% of the constrained combined total, 2.3–5.5% in the other extractions and at most 0.24 of their total uncertainty. Each alternative map moves only 4 to 19 of the pseudo-data events in or out of the signal region, and every shift lies within 1.7 times the Poisson noise of those events, so the term is an upper estimate. The calibrated selection is used from here on.

Table 11: One-bin two-pion total with the MCS-calibrated classifier inputs and the single event classifier, as Table 9 (Table 13 for the two-fold one). The calibration term is not part of the total; added in quadrature it gives 15.4% for the constrained combined total. The unfolded/truth column is one pseudo-data realisation: its excess is a fluctuation of the events migrating across the threshold, not a bias of the calibrated extraction (ensemble means within 1.5%).

	Total	Detector	Flux	Cross sect.	Reint.	Data stat.	Calibration	Unfolded/truth
FHC, no constraint	51.3%	30.9%	36.2%	14.6%	5.1%	10.4%	5.5%	1.082
FHC, constrained	21.3%	10.6%	6.2%	7.8%	5.4%	13.5%	5.1%	1.050
RHC, no constraint	40.3%	19.3%	30.1%	15.0%	5.1%	8.5%	2.5%	1.090
RHC, constrained	15.4%	6.6%	4.2%	5.6%	4.2%	10.7%	2.3%	1.070
Combined, no constraint	44.9%	25.6%	32.5%	14.7%	5.0%	6.6%	3.1%	1.087
Combined, constrained	15.2%	8.0%	4.6%	6.3%	4.7%	8.7%	2.2%	1.056

Ensembles. Pseudo-experiments thrown independently from the nominal calibrated simulation (the eight per-run pseudo-data files each), with the seed of each file a hash of the ensemble, member and file numbers, are extracted through the framework (Figure 8). Over 300 of them the mean extracted total exceeds the expected truth by +0.7 to +1.3%, with a standard error of 0.4 to 0.7%, in FHC, RHC and combined, with and without the constraint. Without the constraint the spread equals the quoted data-statistics term (pull width 1.01 to 1.02; 67 to 72% of the members inside the 68% interval). With it the spread is 0.83 to 0.87 of the quoted term (75 to 79% inside, Figure 9), so the quoted constrained term is conservative; the beam-off, which the framework adds to the pseudo-data without fluctuation while the quoted term counts it, accounts for little of this (without it the constrained data-statistics term of Section 8 is 8.6% instead of 8.8%). A first ensemble of 100 members with consecutive seeds had a spread of 0.85 of the quoted term also without the constraint; the extraction is linear in the signal-region count and its quoted term is the Poisson error of that count, and with hashed seeds the difference disappears. Fifty pseudo-experiments each from GENIE multisim universe 545 and from the $\Delta \to N\pi$ angular variation, extracted with the nominal response, return their own truth within 0.17σ of the total uncertainty (the largest: FHC with the constraint, 0.962 ± 0.014 under the $\Delta \to N\pi$ variation). A stress model outside the covariance, the score-band signal raised by 20% against the signal region, gives 0.910 ± 0.010 of its truth in the combined configuration with the constraint (-0.54σ) and 0.932 ± 0.009 without (-0.14σ): the signal carried by the sidebands moves the constrained result by about half of its total uncertainty for such a change. A second

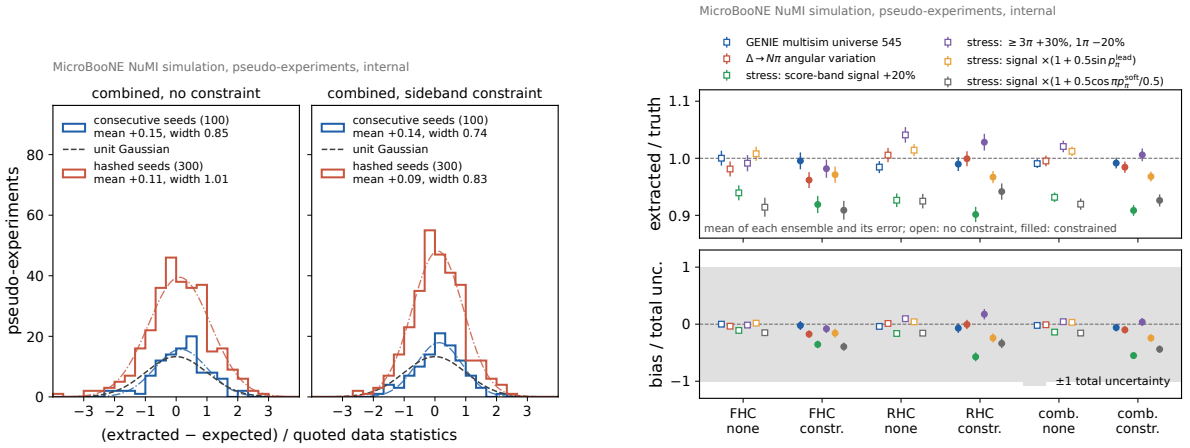


Figure 8: Left: pulls of the nominal ensembles, combined configuration, without and with the constraint (consecutive and hashed seeds). Right: extracted over member truth for the alternative and stress models, per configuration and constraint, with the bias in units of the total uncertainty.

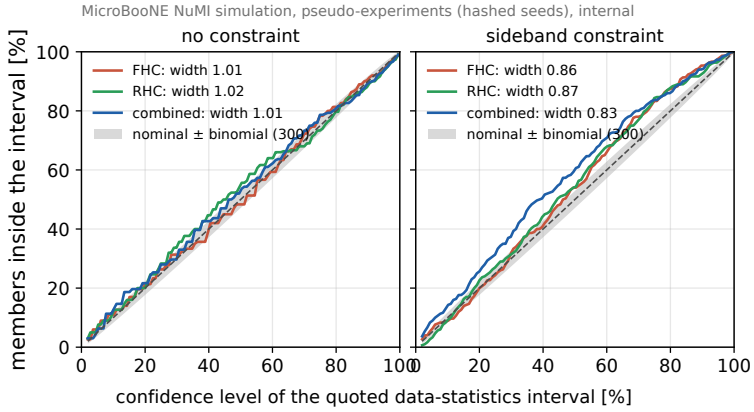


Figure 9: Share of the 300 members of the hashed-seed ensemble inside the quoted data-statistics interval against its confidence level, FHC, RHC and combined, without (left) and with (right) the sideband constraint; the band is the binomial spread of 300 members around the nominal coverage.

stress model, the background with three or more charged pions raised by 30% and the single-pion background lowered by 20%, gives 1.006 ± 0.011 with the constraint and 1.021 ± 0.010 without (within 0.18σ in every configuration). Two signal-shape stress models weight the signal only. The leading-pion weight $1 + 0.5 \sin(p_\pi^{\mathrm{lead}}/\mathrm{GeV})$ raises the signal by 25% almost uniformly over the regions (efficiency +1%): it gives 1.012 ± 0.008 without the constraint and 0.968 ± 0.009 with it (-0.24σ), since the conditioning reads part of the signal excess in the sidebands as background. The softest-pion weight $1 + 0.5 \cos(\pi \min(p_\pi^{\mathrm{soft}}, 0.5\,\mathrm{GeV}/c)/0.5\,\mathrm{GeV}/c)$ lowers the efficiency by 8% and raises the score-band signal against the signal region by 7%: it gives 0.920 ± 0.010 without the constraint and 0.926 ± 0.011 with it (-0.44σ), so a change of the soft-pion spectrum outside the covariance biases the efficiency correction with or without the constraint. The same pseudo-experiments extracted with the reduced-transfer configuration of the framework, without the score band as a sideband, carry a constrained combined total of 20.4% instead of 15.9% (FHC 25.2%, RHC 22.2%; the study gave 19.0%), with a mean offset of −0.4% and a pull width of 0.68; the score-band stress moves the constrained result to 0.939 ± 0.010 (-0.30σ, against -0.55σ with the band) and the softest-pion stress to 0.922 ± 0.012 (-0.36σ): leaving the band out halves the transfer of the signal model and leaves the efficiency effect as it is.

Selection variants. The working point was scanned again on the constrained combined total (score above 0.85 to 0.96) with the evaluation of Section 8 outside the framework (the framework gives 15.2% at the same threshold): 0.92 remains the minimum (16.8%; Figure 3, right), the total changes little between 0.90 and 0.94, and tighter thresholds reach 58–70% purity at a total of 22–32%. Per horn mode the minimum lies at 0.90 in FHC (25.9%; 27.3% at 0.92, within 2.7 points from 0.85 to 0.95) and at 0.92 in RHC (19.3%). Event-classifier features from the reconstructed showers (energy, vertex distance, track score and direction of the leading shower; with its dE/dx, the second shower and the two-shower mass in a larger set) raise the area under the ROC curve on the validation periods from 0.874 to 0.884–0.887 and the purity at a fixed threshold by about two points. In the π^0 control region, against its own normalisation, the energy, vertex distance, track score and direction of the leading shower follow the beam-on data; its dE/dx and the multi-shower features do not (χ^2 of 22–63 for 3 to 6 degrees of freedom). With the calibrated inputs the best constrained totals of the variants, with and without shower features and with the calorimetric particle classifier, are 16.3–17.3% (21.1% for the calorimetric classifier without shower features), against 16.8% for the adopted selection; the detector term of one variant moves by 3–6 points between neighbouring thresholds, which is more than these differences. The selection is kept.

11 Final selection: the two-fold event classifier

Overtraining. Over all run periods the signal region of the selection above holds 395 predicted signal events, against 378 projected from the validation periods (Table 7): the selection efficiency is $14.04 \pm 0.21\%$ in the run periods on which the classifiers were trained and $13.16 \pm 0.31\%$ in the validation periods (RHC Run 3 $13.60 \pm 0.37\%$, FHC Run 5, a separate production, $12.17 \pm 0.61\%$), a difference of 2.4σ. It is overtraining of the event classifier, which was trained on a random 80% of the events of the training periods: within the domain, at a score above 0.92, its efficiency is $23.61 \pm 0.36\%$ on those events, $21.35 \pm 0.71\%$ on the other 20% of the same periods and $21.18 \pm 0.48\%$ in the validation periods (purity 55.4, 51.0 and 51.4%). The held-out events of the training periods behave as the validation periods, so the particle classifier adds no measurable effect. A response built from all periods overestimates the efficiency of independent events, and of the data, by about 6% and underestimates their background by about 4%, about -2% on the unconstrained total; closure and ensembles cannot see it, since their pseudo-data come from the same events.

Two folds. The event classifier is therefore trained twice, on the events with even and with odd event numbers, and every event, data and beam-off included, is scored by the model of the other parity, so that no simulated event is scored by a model that saw it. The selection reads the event number of the ntuple and holds both models; on a 30 000-event slice its scores equal those of the study to 6×10^{-8}. Each fold model is overtrained on its own events (efficiency at 0.92 21.3 against 19.5%, 19.7 against 18.8%), but no event is scored by it. With the out-of-fold scores the constrained combined total of the study is 17.4, 20.4, 16.5, 19.5 and 25.5% at 0.85, 0.88, 0.90, 0.92 and 0.94, the scatter between neighbouring thresholds again set by the detector term, and the working point moves to its minimum, 0.90. Table 12 gives the regions. In the study the constraint takes the combined total from 50.1% to 16.5%; with the sideband signal decoupled it is 19.6%, without the score band 21.2% and with both 23.9%, and the two reweighted models close within 0.21σ.

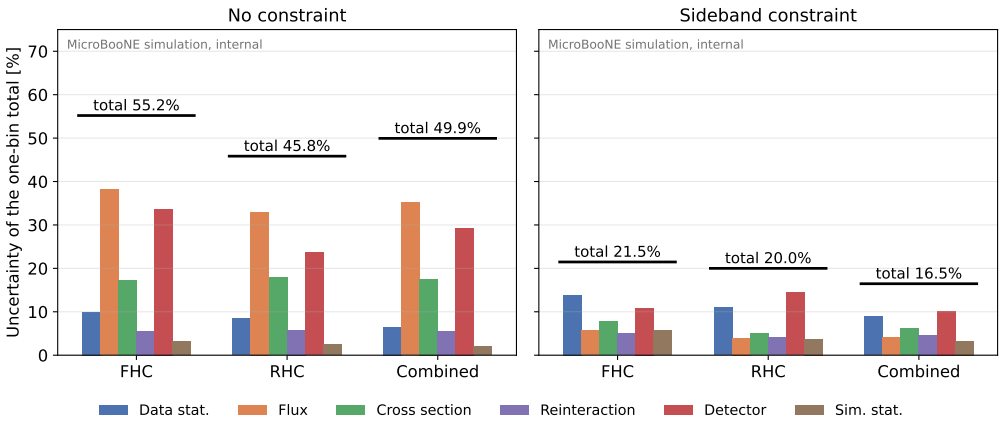


Figure 10: Terms of the uncertainty of the one-bin total with the two-fold event classifier and the MCS-calibrated classifier inputs (Table 13), without and with the sideband constraint.

Table 12: Signal region and sidebands with the two-fold event classifier (working point 0.90), combined configuration, predicted events at full exposure over all run periods (every event scored by the model of the other parity).

Region	Events	Signal	Purity	Largest classes
Signal region (score > 0.90)	920	444	48%	CC1π 16%, CC$\geq 3\pi$ 13%
Three-pion-like	294	36	12%	CC$\geq 3\pi$ 24%, beam-off 22%, CCπ^0 16%
Proton-like	16 640	458	3%	CC0π 37%, CC1π 18%, CCπ^0 14%
Score 0.5–0.90	4 534	655	14%	CCπ^0 24%, CC1π 24%
Score < 0.5	13 162	166	1%	beam-off 54%, CC0π 9%, CCπ^0 8%

Extraction. The 46 samples were processed again with the two-fold classifier (and the calibrated MCS inputs and the differential variables) and the pseudo-data thrown with the same seeds, so that they hold the same events as before. Table 13 and Figure 10 give the one-bin total. The constrained combined total is 16.5% (15.2% with the single model), 17.1% with the MCS-calibration term evaluated again for the two-fold classifier (4.7%; 5.9% in FHC and in RHC, 0.29 of the total; each map moves 9 to 26 pseudo-data events in or out of the signal region, the shifts within 3.1 times their Poisson noise, so the term is in part noise and an upper estimate); the RHC detector term moves from 6.6% to 14.4%, within the scatter of that term between selections. A pseudo-experiment ensemble of 200 members with hashed seeds returns the expected truth within +0.0 to +1.0% (standard error 0.5 to 0.8%) with and without the constraint, with pull widths 0.96 to 1.01 without the constraint and 0.81 to 0.85 with it, as for the single model (Figure 12). The constrained data-statistics term is conservative because it propagates the data covariance with the gain of the conditioning of the whole signal-region prediction, signal included, while the estimator subtracts the background with the gain of the background alone, which is smaller because the sidebands contain signal: in the study the two give 8.6% and 7.6% (ratio 0.88; 0.77 to 0.99 for single sidebands), and with the unfluctuated beam-off (0.98) this accounts for the measured pull widths. The framework term is kept as the nominal data-statistics uncertainty, as for every other term. Fifty pseudo-experiments each from GENIE multisim universe 545 and from the $\Delta \to N\pi$ angular variation return their own truth within 0.11σ; the score-band stress model gives 0.915 ± 0.010 in the combined configuration with the constraint (-0.47σ; 0.946 without) and the softest-pion stress model 0.943 ± 0.012 (-0.32σ; 0.934 without), as for the single model. In the opened control regions, with the skim processed again, the event score of the two-fold classifier gives $\chi^2 = 7.4/11$ with the systematic covariance (80.8/11 with statistics alone), the

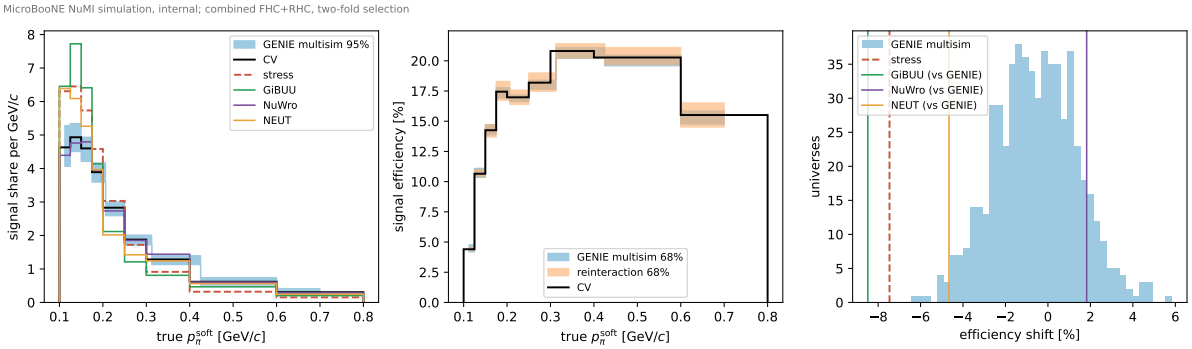


Figure 11: Softest pion of the two-pion signal, combined configuration, two-fold selection. Left: true spectrum of the CV, the GENIE multisim universes (95%), the stress model and the standalone generators. Middle: signal efficiency with the multisim and reinteraction universes (68%). Right: efficiency shift of the multisim universes, of the stress model and of the generators against GENIE.

summed $P(\pi)$ 2.0/7, and the LLR score of the pool tracks 20.3/9 ($p = 0.016$), as in Section 10. With the recombination shape as a free strength and every other term in the covariance, the data call for 1.44 ± 0.50 times the simulated variation (χ^2 26.8 without it, 18.7 fitted, 8 degrees of freedom); at that size it moves the constrained combined total by 0.27 of its total uncertainty, the FHC total by 0.40 and the differential integrals by at most 0.31 (0.40 in a single bin), of which the part beyond the recombination term of the covariance is at most 0.12. This is the selection of the analysis; the tables of Sections 6 to 10 document the single model it replaces.

Soft pions. The signal efficiency rises from 4% for a softest pion at 0.10 to 0.125 GeV/c to 14% at 0.15 to 0.175 GeV/c, is 17 to 21% from there to 0.6 GeV/c and 15% above (Figure 11). The 600 GENIE universes put 30 to 40% of the signal below 0.175 GeV/c (CV 35%) and move the efficiency through the spectrum by 1.2% (standard deviation; 2.0% in all, 3.1% for the reinteraction universes), while the softest-pion stress model (46% below 0.175 GeV/c) moves it by -7.4%, 3.8 times the standard deviation of all universes. In the standalone generators 35% (NuWro), 37% (GENIE), 44% (NEUT) and 52% (GiBUU) of the signal lie below 0.175 GeV/c; reweighting the signal by the spectrum of each over that of GENIE shifts the efficiency by $+1.8$, -4.4 and -8.3%. The stress model lies inside the range of the generators and outside the covariance. Restricting the signal to pions above 0.175 GeV/c does not remove the dependence, since the softer events stay in the signal region as background; the events with the softer pion below 0.10 GeV/c change by at most 0.9% of the signal. A soft-pion efficiency term, the largest of these shifts, is therefore added: 8.5, 8.1 and 8.3% in FHC, RHC and combined, which takes the constrained totals to 23.1, 21.6 and 18.4%; per bin of the differential variables, propagated through the unfolding, it is 6 to 12% (Section 12). It is applied in full with and without the constraint.

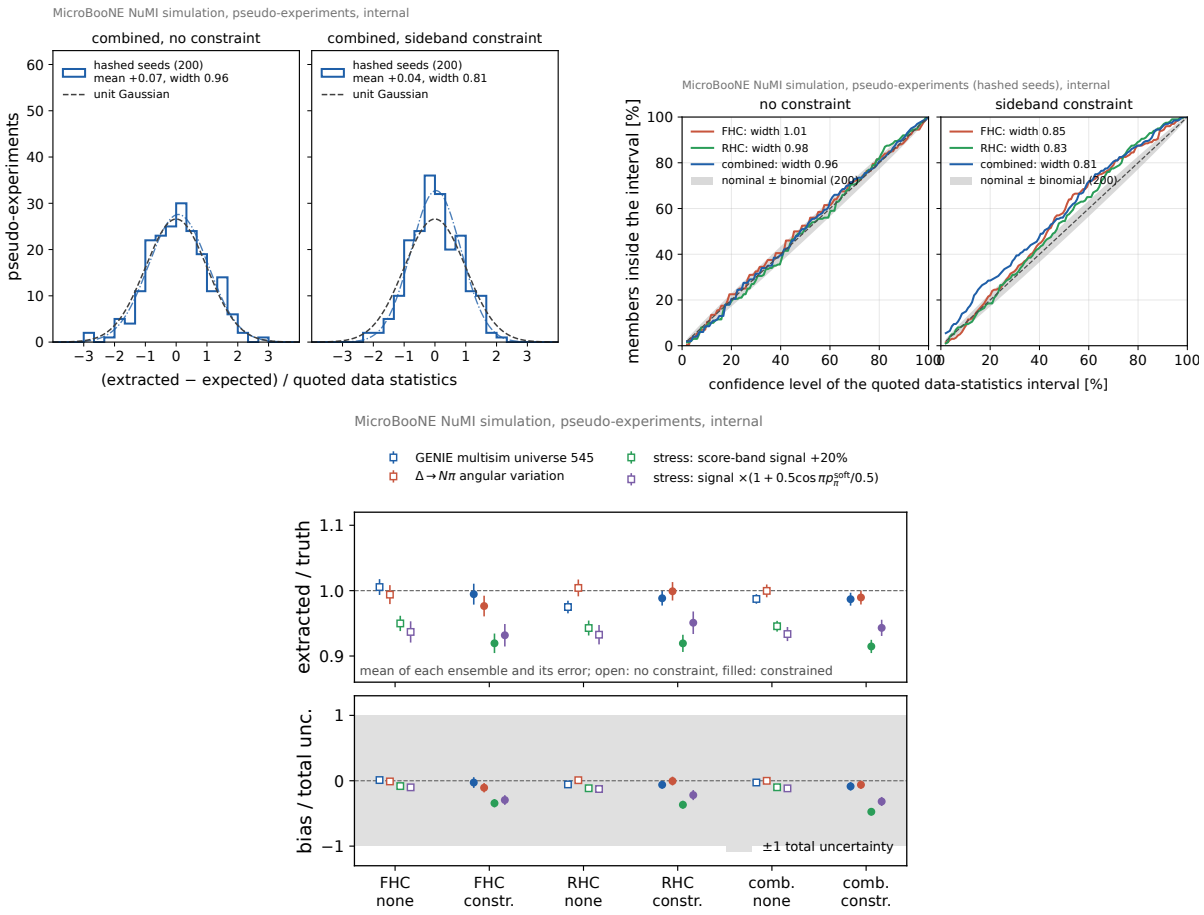


Figure 12: Ensembles with the two-fold event classifier: pulls of the 200 nominal members in the combined configuration (top left), the share of members inside the quoted data-statistics interval against its confidence level (top right), without and with the sideband constraint, and extracted over member truth for the reweighted and stress models (bottom).

Table 13: One-bin two-pion total with the two-fold event classifier at the working point 0.90, MCS-calibrated classifier inputs, through the framework on pseudo-data, as Table 11. The unfolded/truth column is one pseudo-data realisation; the ensemble means are within 1%. With the soft-pion efficiency term the constrained totals are 23.1, 21.6 and 18.4%, with the MCS-calibration term as well 23.8, 22.4 and 19.0%.

	Total	Detector	Flux	Cross sect.	Reint.	Data stat.	Unfolded/truth
FHC, no constraint	55.2%	33.7%	38.2%	17.2%	5.6%	10.0%	1.125
FHC, constrained	21.5%	10.9%	5.7%	7.7%	5.0%	13.9%	1.068
RHC, no constraint	45.8%	23.7%	33.0%	17.9%	5.8%	8.5%	1.071
RHC, constrained	20.0%	14.4%	3.9%	4.9%	4.1%	11.0%	1.059
Combined, no constraint	49.9%	29.3%	35.1%	17.5%	5.6%	6.5%	1.093
Combined, constrained	16.5%	10.2%	4.0%	6.1%	4.7%	8.9%	1.059

12 Differential variables

The differential measurements use the tracks of the assignment (Section 6). Table 14 defines the variables. As in the single-pion analysis, reconstructed angles are taken about the direction from the NuMI target to the reconstructed vertex and true angles about the true neutrino direction, and the muon momentum is the range momentum when the track ends inside the containment volume and the MCS momentum otherwise, calibrated for simulated tracks with the map of Section 10. The two pions are ordered by the variable itself (the more forward and the more backward pion, the higher and the lower momentum), the same way at truth and reconstructed level, so that the convention is symmetric in the pions and needs no pion identity. The pion momentum is the range momentum of a contained track, converted from the muon hypothesis at equal kinetic energy over mass, and is defined only when both assigned pions are contained. The proton multiplicity counts, at reconstructed level, the tracks outside the assignment with $P_p > 0.5$, and at truth level the protons above 0.3 GeV/c of the subsample of D1.

Table 14: Variables of the differential measurements. Reconstructed quantities come from the assigned tracks.

Variable	Reconstructed	True
p_μ	range (contained end) or calibrated MCS	muon momentum
$\cos\theta_\mu$	about the target-to-vertex direction	about the neutrino direction
$\cos\theta_\pi^{\mathrm{fwd}}$, $\cos\theta_\pi^{\mathrm{bwd}}$	the larger and the smaller of the two pion cosines	the same for the true pions
$\cos\theta_{\pi\pi}$	opening angle of the assigned pions	of the true pions
$\cos\theta_{\mu\pi}$	muon and the closer assigned pion	muon and the closer true pion
p_π^{low}, p_π^{high}	range momenta, both pions contained	true pion momenta
N_p	unassigned tracks with $P_p > 0.5$	protons above 0.3 GeV/c

The binnings are contiguous, each bin with $P(\text{reconstructed bin } i \,|\, \text{true bin } i) \geq 0.50$, the migration criterion of the single-pion release, and at least 40 predicted signal events in the signal region of the combined configuration at the full exposure, chosen with the single classifier. Table 15 gives them evaluated with the two-fold classifier (every diagonal at least 0.50, at least 44 signal events per bin), with the stability $P(r_i|t_i)$ and the reconstructed-bin purity $P(t_i|r_i)$ of the signal, the efficiency, the purity, flat at 43 to 54%, and $\sqrt{N}/S$, the data statistics of the background-subtracted count of a bin before unfolding and constraint, and Figures 15 to 17 the predicted events by true class, the response of the signal and the efficiency. The muon variables and the pion and opening angles pass with 5 to 8 bins at 12 to 23% per bin, and the efficiency varies little across them (12 to 17%); the reconstructed-bin purity is lowest in the upper p_μ bins

(0.30 to 0.35) and the 0-proton bin (0.36), where events of neighbouring true bins feed in, which the unfolding corrects through the response. Both assigned pions are contained in 79% of the selected signal; the other events have no reconstructed pion momentum and enter the response of the pion momenta as inefficiency, hence their lower efficiency. The range momentum saturates as in the single-pion analysis, since most pions interact before they stop: the lower and the higher pion momentum keep 3 and 2 bins, and with the bins of D2, above 0.175 GeV/c, the lower one keeps 2. The proton multiplicity has two bins, 0 and at least one proton, the second comparable with the BNB result. The selection writes these variables for every event (reconstructed for the assignment, true for the signal); on a 30 000-event slice they equal the values of this study to 10^{-7} and every other branch is unchanged. The bin configurations carry the signal-region bins in the variable and the four sidebands of the one-bin total, so that the constraint acts as for the total.

Table 15: Binnings of the differential variables under the 0.50 criterion, evaluated with the two-fold event classifier (signal region 920 events, 444 signal, Table 12), combined configuration, full exposure: stability $P(r_i \,|\, t_i)$ and reconstructed-bin purity of the signal $P(t_i \,|\, r_i)$, efficiency, purity and $\sqrt{N}/S$ per bin (ranges over the bins). Open last bins of the momenta are marked ∞.

Variable	Bins	Edges	$P(r\|t)$	$P(t\|r)$	Efficiency	Purity	$\sqrt{N}/S$
p_μ [GeV/c]	5	0.15, 0.35, 0.5, 0.65, 0.9, ∞	0.50–0.68	0.30–0.95	11.5–16.7%	47–51%	12–20%
$\cos\theta_\mu$	6	-1, 0.35, 0.65, 0.8, 0.9, 0.95, 1	0.73–0.91	0.72–0.82	12.0–17.3%	44–50%	13–23%
$\cos\theta_\pi^{\mathrm{fwd}}$	6	-1, 0.45, 0.7, 0.8, 0.9, 0.95, 1	0.61–0.78	0.56–0.82	12.7–16.4%	47–51%	13–20%
$\cos\theta_\pi^{\mathrm{bwd}}$	8	-1, -0.65, -0.3, 0, 0.25, 0.45, 0.6, 0.75, 1	0.64–0.83	0.62–0.83	13.7–16.1%	43–54%	16–22%
$\cos\theta_{\pi\pi}$	8	-1, -0.8, -0.45, -0.1, 0.2, 0.45, 0.65, 0.8, 1	0.53–0.71	0.53–0.82	13.8–17.4%	45–53%	16–22%
$\cos\theta_{\mu\pi}$	7	-1, 0.15, 0.4, 0.55, 0.65, 0.75, 0.85, 1	0.53–0.79	0.49–0.77	12.6–17.4%	46–51%	14–21%
p_π^{low} [GeV/c]	3	0.10, 0.175, 0.225, ∞	0.56–0.63	0.35–0.84	8.3–14.3%	45–51%	13–15%
p_π^{high} [GeV/c]	2	0.10, 0.275, ∞	0.74–0.75	0.30–0.95	9.6–12.7%	45–50%	9–14%
N_p	2	0, ≥ 1	0.73–0.92	0.36–0.98	11.6–16.3%	43–52%	8–12%

Extraction through the framework. The muon variables, the four angles and the proton multiplicity were extracted through the framework with the two-fold event classifier (Section 11) in FHC, RHC and combined, without and with the sideband constraint, with Wiener-SVD unfolding (filter on, second-derivative regularisation, as the single-pion differential results) on the pseudo-data of the one-bin total (Table 17, Figure 14); the bins are those of Table 15. In the nominal pseudo-data realisation every extraction is compatible with its truth smeared by A_C, the regularisation matrix of the unfolding, applied to every prediction compared with an unfolded result: the χ^2 with the total covariance is at most 4.1 for 7 bins (combined) and 4.6/7 in a single horn mode. The constraint sets the precision of the differential results: without it the per-bin uncertainty is 40 to 187% in the combined configuration, with it 17 to 59%, the edge bins with the lowest efficiency or the largest migration being the least precise. These uncertainties contain the soft-pion efficiency term (Section 11): a change δ of the efficiency per true bin moves the unfolded result by $A_C(\delta\, t)$, t the tune truth, and the term is that shift for the envelope of the generators (in every bin the largest), fully correlated over the bins, 6 to 12% per bin. The bin-averaged uncertainty is 23 to 40%; the integral over the bins, with the full covariance, carries 16.9 to 19.6% (data statistics 9.0 to 9.9%), as the one-bin total with the same term (18.4%, 8.9%), and lies 2 to 12% below it, the effect of the A_C smearing of the regularised unfolding on the integral; the one-bin total remains the normalisation result. For N_p a plain inversion (Wiener filter off) gives the same integral uncertainty (17.9%) and a larger one in the 0-proton bin (86% against 58%).

Ensembles. The 200 members of the one-bin ensemble (Section 11) were extracted for every variable in the combined configuration (Figure 13). They return their own A_C-smeared truth within 0.11 of the total uncertainty in every bin and their integrals within −1.1 to +2.4%. Against the expected truth, with the data-statistics covariance, the pulls per bin have widths 0.89 to 1.04 without the constraint and 0.83 to 1.06 with it, and the mean χ^2 is 0.95 to 1.00 times the number of bins without the constraint and 0.81 to 0.96 times with it, as for the one-bin total. The closure χ^2 of the framework adds the Poisson variance of the true count to the unfolded covariance, which already holds the data statistics, so its values are small; against its distribution over the members the nominal realisation has $p = 0.42$ to 0.89 for six variables and 0.03 (without the constraint) and 0.05 (with it) for $\cos\theta_{\mu\pi}$. Fifty members each from GENIE multisim universe 545 and from the $\Delta \to N\pi$ angular variation return their truth within 0.43 and 0.20 of the total uncertainty per bin (integrals within 9.2 and 2.9%, the largest for the backward pion without the constraint). The softest-pion stress model moves single bins by up to 16%, 0.53 of the total uncertainty and up to 1.9 times the soft-pion term of the bin, and the integrals by 4 to 13% without and 4.5 to 6.2% with the constraint.

Generator predictions. GENIE v3 (G18_10a_02_11a, untuned), GiBUU, NuWro and NEUT predict the two-pion signal from the standalone samples of the single-pion analysis, with its signal definition and the true variables of the selection, normalised per nucleon as there and combined over ν_μ and $\bar{\nu}_\mu$ by flux. These samples were generated with the FHC flux of each flavour; for RHC every event is weighted by the ratio of the RHC to the FHC flux shape at its energy, since the reverse-horn ν_μ flux is harder (mean 0.54 against 0.47 GeV) and the $\bar{\nu}_\mu$ flux softer, and the two-pion cross section rises with energy: unweighted, RHC would be 20% low for GENIE. With the weight GENIE gives RHC/FHC = 1.01, as the overlay (1.02). The generators predict 0.61 (NuWro) to 0.86 (NEUT) of the combined pseudo-data total, which follows the tune (−2.0 to −0.8 times its uncertainty with the soft-pion and MCS-calibration terms), and differ in shape mainly in the muon and pion angles (Table 16, Figure 14). The pseudo-data come from the tune, so these comparisons show the sensitivity of the measurement, not a test of the models.

Table 16: χ^2 of the A_C-smeared generator predictions against the unfolded pseudo-data, combined configuration with the sideband constraint, covariance including the soft-pion term, and the ratio of the predicted to the unfolded integral.

Variable	Bins	GENIE	GiBUU	NEUT	NuWro
p_μ	5	3.0, 0.75	5.3, 0.66	1.4, 0.86	6.1, 0.61
$\cos\theta_\mu$	6	3.6, 0.74	6.4, 0.65	3.6, 0.85	6.9, 0.60
$\cos\theta_\pi^{\mathrm{fwd}}$	6	3.8, 0.74	5.0, 0.66	2.3, 0.85	8.6, 0.60
$\cos\theta_\pi^{\mathrm{bwd}}$	8	3.4, 0.78	7.6, 0.73	1.4, 0.89	5.6, 0.64
$\cos\theta_{\pi\pi}$	8	3.8, 0.76	7.9, 0.71	3.7, 0.90	8.1, 0.65
$\cos\theta_{\mu\pi}$	7	5.8, 0.77	7.4, 0.69	6.4, 0.89	7.3, 0.62
N_p	2	1.8, 0.75	4.0, 0.66	0.8, 0.85	4.7, 0.61

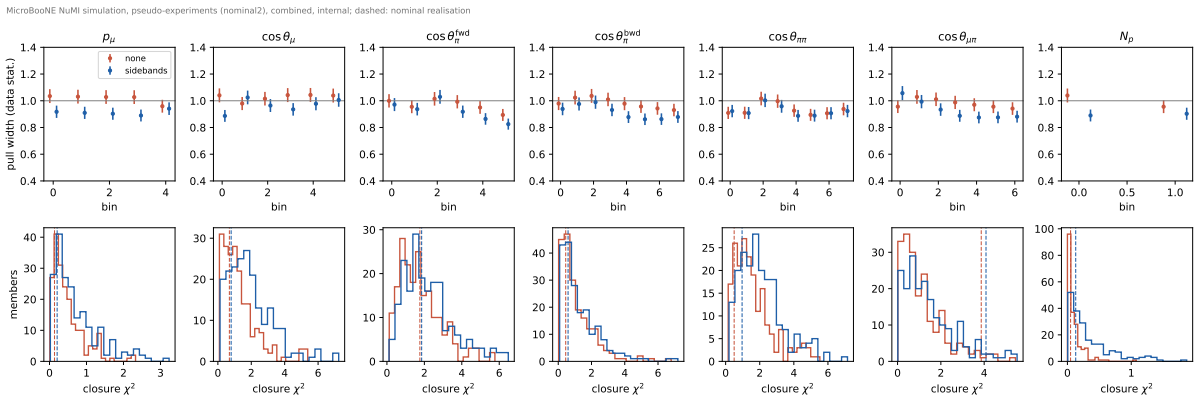


Figure 13: Ensembles of the differential extractions, combined configuration, 200 members: width of the pulls against the expected truth with the data-statistics covariance per bin (top) and the closure χ^2 of the framework over the members, with the nominal realisation dashed (bottom), without and with the sideband constraint.

Table 17: Differential extractions through the framework with the two-fold event classifier, combined configuration, nominal pseudo-data realisation: total uncertainty per bin, soft-pion efficiency term included (range over the bins), without and with the sideband constraint, the bin average (mean of the fractional bin uncertainties) and the uncertainty of the integral over the bins with its data statistics, from the full covariance (constrained), and the χ^2 of the unfolded pseudo-data against their A_C-smeared truth and against the tune with the covariance of the framework (constrained, without the soft-pion term), for the number of bins; p is the share of the 200 pseudo-experiments with a closure χ^2 at or above the nominal one.

Variable	Bins	Per bin, none	Per bin, constr.	Bin average	Integral (data stat.)	χ^2 truth (p)	χ^2 tune
p_μ	5	45–61%	19–34%	28.3%	17.7% (9.2%)	0.21/5 (0.84)	0.57/5
$\cos\theta_\mu$	6	41–124%	18–54%	34.4%	17.6% (9.1%)	0.79/6 (0.81)	1.43/6
$\cos\theta_\pi^{\mathrm{fwd}}$	6	51–114%	23–53%	35.5%	16.9% (9.0%)	1.85/6 (0.51)	1.68/6
$\cos\theta_\pi^{\mathrm{bwd}}$	8	46–100%	17–43%	23.0%	18.3% (9.9%)	0.59/8 (0.69)	1.79/8
$\cos\theta_{\pi\pi}$	8	43–67%	19–32%	24.8%	17.0% (9.9%)	0.96/8 (0.82)	1.83/8
$\cos\theta_{\mu\pi}$	7	40–187%	19–47%	26.3%	18.3% (9.1%)	4.08/7 (0.05)	5.30/7
N_p	2	44–130%	22–59%	40.3%	19.6% (9.9%)	0.13/2 (0.67)	0.08/2

13 Three or more pions

In the two-pion domain, events with at least four tracks in which three pions are more likely than two number 234 at full exposure (combined, with the deployed particle classifier of the Phase-2 study; the three-pion-like sideband of the adopted selection holds 294, Table 12), of which 88 have three or more charged pions, 41 a neutral pion and 35 are two-pion signal. A three-pion event classifier built the same way reaches a ROC area of 0.855, but with all terms every three-pion selection carries a total uncertainty above 100% (`CC1mu3pi` 245%, with the particle classifier 117%, the event classifier at best 105%). Four and five pions are out of reach: combined, 3.1 selected four-pion signal events at 4.8% purity and 0.1 five-pion events, and the probability that a true pion multiplicity is reconstructed in its own bin is 0.84, 0.48 and 0.33 for one, two and three or more pions. The three-pion-like events are therefore the control sample of Section 8 (D8).

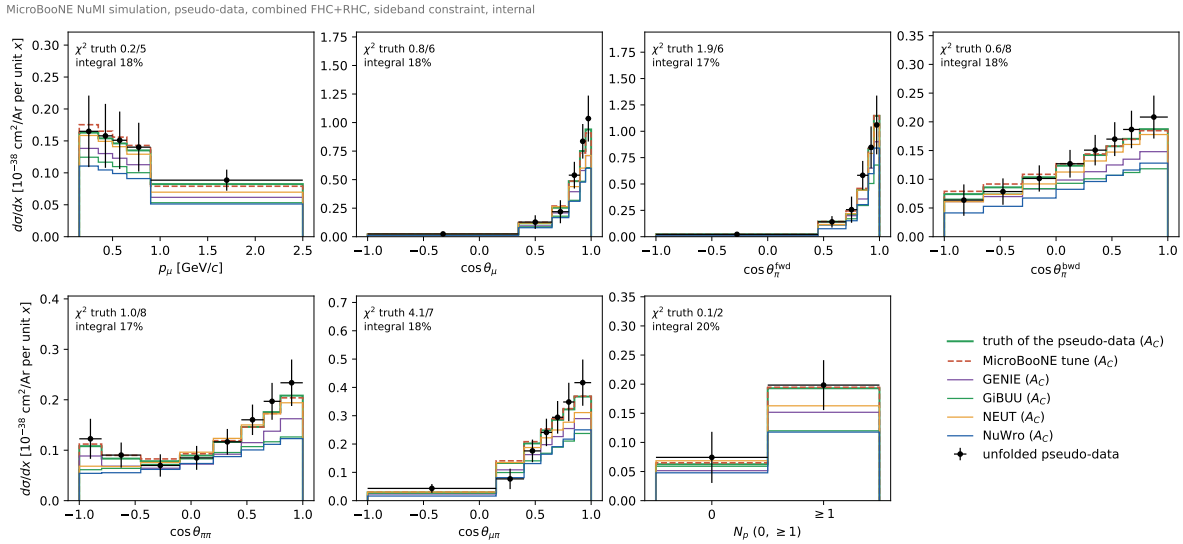


Figure 14: Differential results on pseudo-data with the two-fold event classifier, combined configuration with the sideband constraint: unfolded pseudo-data with the total uncertainty (soft-pion term included), the A_C-smeared truth of the pseudo-data, the A_C-smeared tune and the A_C-smeared generator predictions.

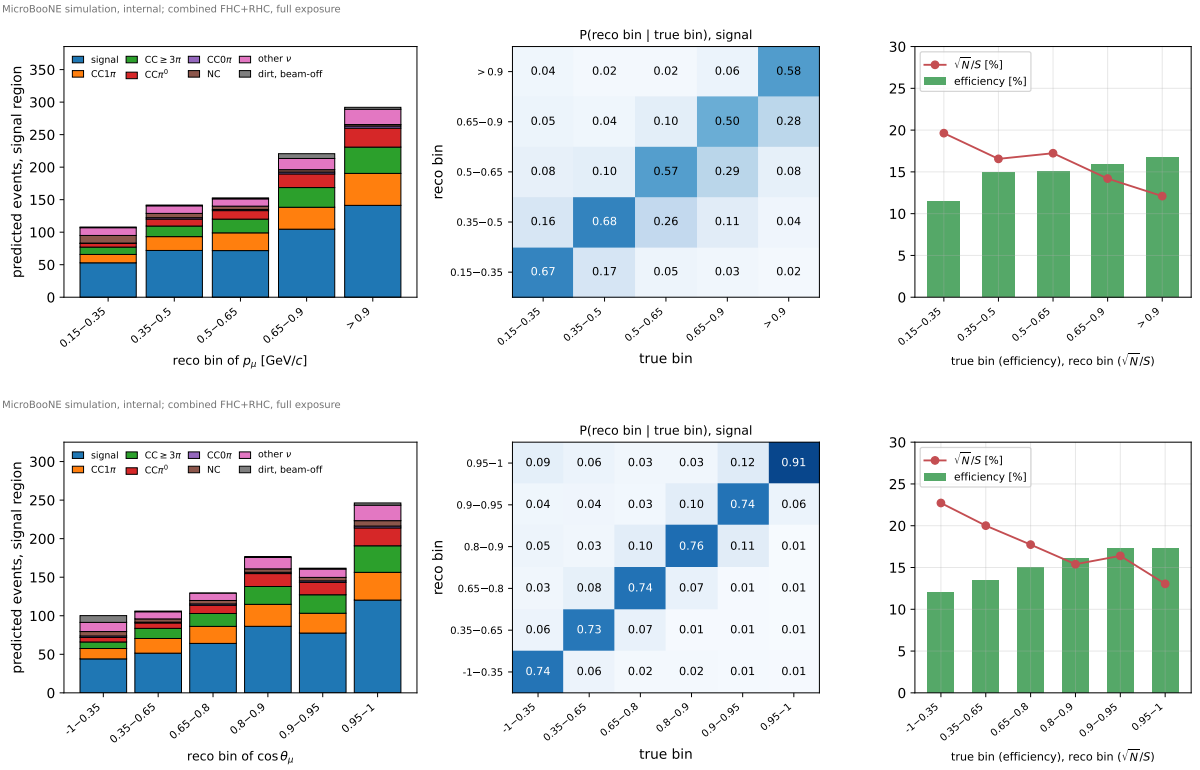


Figure 15: Muon momentum (top) and angle (bottom), two-fold event classifier: predicted events of the signal region by true class in the bins of Table 15 (left), $P(\text{reconstructed bin} \,|\, \text{true bin})$ of the signal (centre), efficiency per true bin and $\sqrt{N}/S$ per reconstructed bin (right); combined configuration, full exposure.

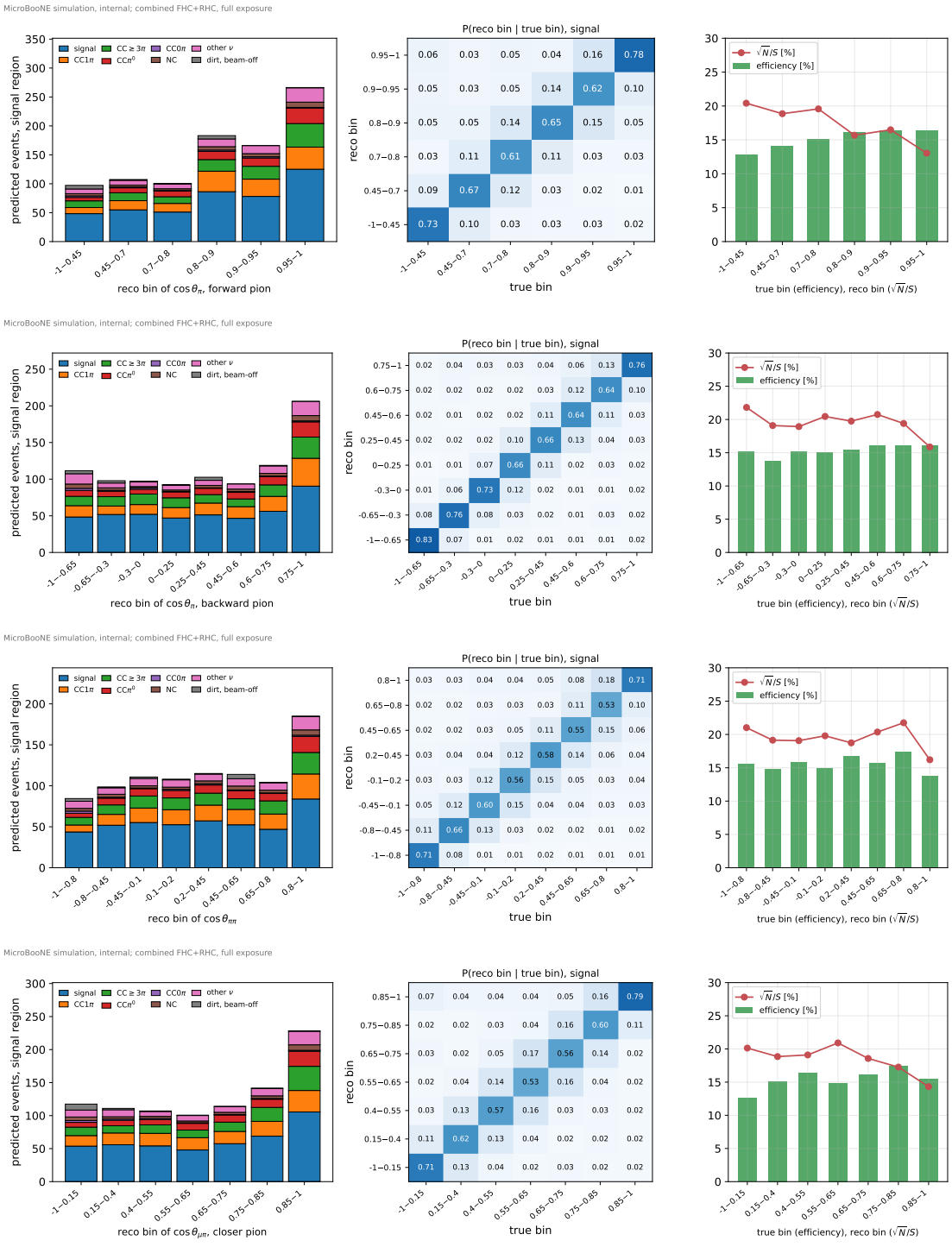


Figure 16: As Figure 15, for the angles of the forward and the backward pion, the opening angle of the two pions and the opening angle of the muon and the closer pion (top to bottom).

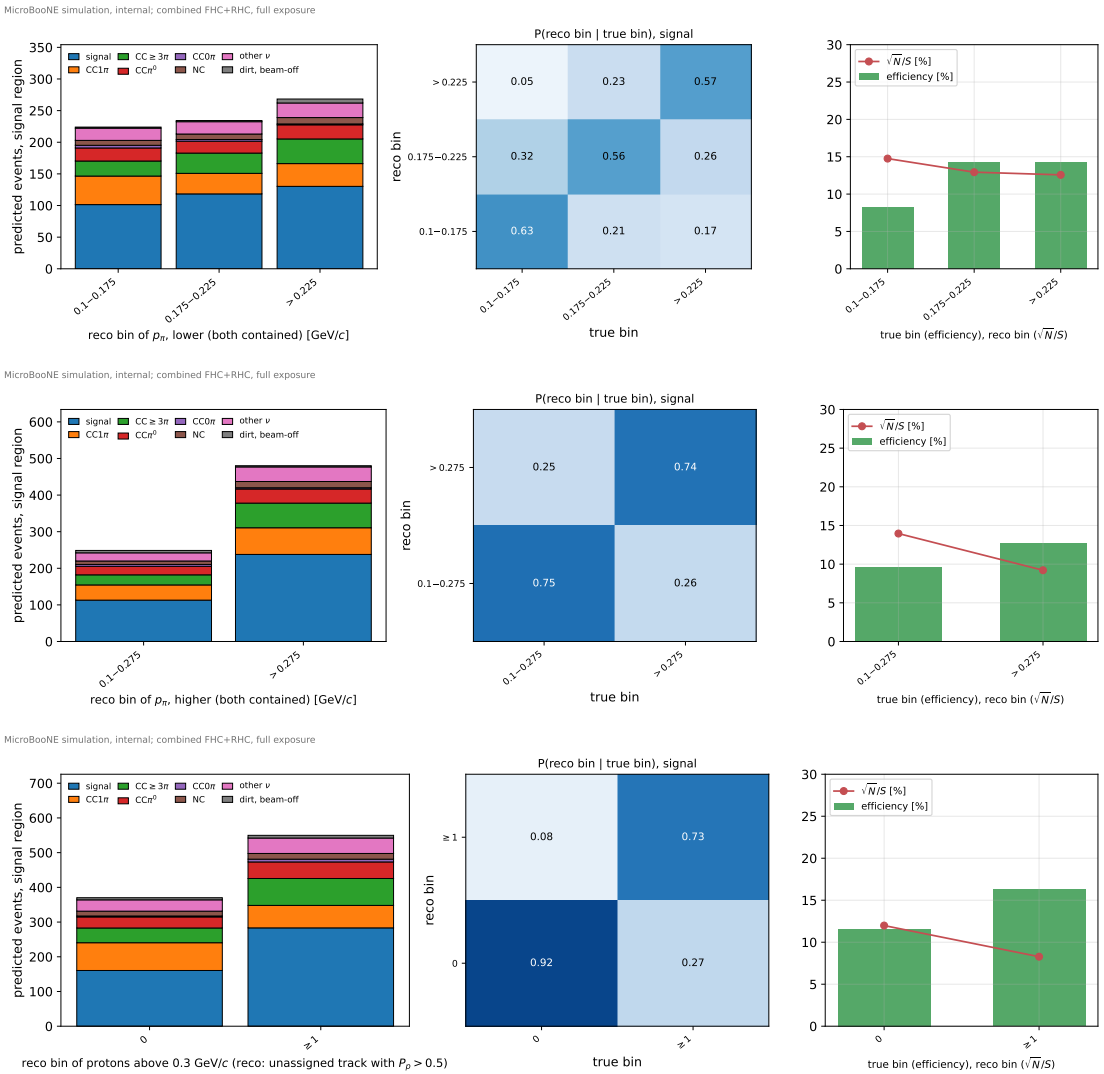


Figure 17: As Figure 15, for the lower and the higher pion momentum (both pions contained) and the proton multiplicity (top to bottom).

14 Status and next steps

- The two-pion one-bin total is extracted through the framework in FHC, RHC and combined, with and without the sideband constraint, with the MCS-calibrated classifier inputs and the two-fold event classifier (Table 13).
- The inputs of the LLR-and-geometry classifier and the event score are compared with beam-on data in the opened control regions (Section 10): the MCS momentum is calibrated, and the LLR of long protons follows the recombination variation of the detector term.
- The selection was compared with event-classifier features from the reconstructed showers and with the calorimetric particle classifier on the constrained total (Section 10); no variant improves it beyond the statistics of the detector-variation samples, and the selection is kept.
- Validation through the framework: ensembles of pseudo-experiments, alternative-model and stress-model ensembles (Sections 10 and 11).
- The constrained data-statistics term is the framework's, conservative against the ensembles; the LLR score of the pool tracks calls for the recombination variation at 1.4 times its size, which moves no result by more than 0.12 of its uncertainty beyond the detector term (Section 11).
- The softest-pion spectrum is not covered by the interaction-model covariance: a soft-pion efficiency term, the envelope of the standalone generators (8.3% combined, 6 to 12% per bin in the covariance of the differential results), is added; the migration across the 0.10 GeV/c threshold is below 1% (Section 11).
- Generator predictions with the two-pion signal definition (GENIE, GiBUU, NuWro, NEUT) enter the extractions; the RHC predictions are weighted to the RHC flux shape (Section 12). The single-pion generator predictions use the FHC-flux samples for RHC unchanged.
- The overtraining of the single event classifier (about 6% on the signal efficiency) is removed by training it in two folds by event-number parity, with the working point at 0.90 (Section 11).
- The differential variables of Section 12 are extracted through the framework with the sideband constraint, are compatible with their truth in the nominal pseudo-data realisation and close in nominal and alternative-model ensembles. Open for them: the pion momenta; the invariant masses remain study products unless they pass the 0.50 criterion.
- Perform an independent-generator closure if a NuMI NuWro overlay can be validated (D5).
- Apply and validate the same sideband-constraint method in the single-pion analysis using pseudo-data.

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